Database Laboratory Report

Course Code : CO313

Submitted By

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Course : BTech 5th Sem

Roll No : CSB23017

Q1. List each employee's full name (fname, lname) along with their department name.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT e.fname, e.lname, d.name AS dept_name
    -> FROM EMPLOYEE e
    -> JOIN DEPARTMENT d ON e.dept_no = d.dept_no;
+--------+--------+----------------+
| fname  | lname  | dept_name      |
+--------+--------+----------------+
| Ramen  | Kalita | Administration |
| Rajesh | Gogoi  | Administration |
| Priya  | Baruah | Headquarters   |
| Arun   | Sarma  | Research       |
| Anjali | Das    | Research       |
| Deepa  | Bora   | Research       |
+--------+--------+----------------+
6 rows in set (0.00 sec)
```

Q2. Retrieve the project number and name of all projects controlled by the department named 'Research'.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT p.proj_no, p.name
    -> FROM PROJECT p
    -> JOIN DEPARTMENT d ON p.dept_no = d.dept_no
    -> WHERE d.name = 'Research';
+---------+----------+
| proj_no | name     |
+---------+----------+
|      10 | ProductX |
|      20 | ProductY |
+---------+----------+
2 rows in set (0.00 sec)
```

Q3. For each department, show dept_no, department name, and the number of employees in that department.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT d.dept_no, d.name, COUNT(e.ssn) AS num_employees
    -> FROM DEPARTMENT d
    -> LEFT JOIN EMPLOYEE e ON d.dept_no = e.dept_no
    -> GROUP BY d.dept_no, d.name;
+---------+----------------+---------------+
| dept_no | name           | num_employees |
+---------+----------------+---------------+
|       2 | Administration |             2 |
|       3 | Headquarters   |             1 |
|       1 | Research       |             3 |
+---------+----------------+---------------+
3 rows in set (0.00 sec)
```

Q4. List the ssn, full name, and department of employees who have no supervisor (super_ssn IS NULL).

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT e.ssn, e.fname, e.lname, d.name AS dept_name
    -> FROM EMPLOYEE e
    -> JOIN DEPARTMENT d ON e.dept_no = d.dept_no
    -> WHERE e.super_ssn IS NULL;
+-----+--------+--------+----------------+
| ssn | fname  | lname  | dept_name      |
+-----+--------+--------+----------------+
| 104 | Rajesh | Gogoi  | Administration |
| 103 | Priya  | Baruah | Headquarters   |
| 100 | Arun   | Sarma  | Research       |
+-----+--------+--------+----------------+
3 rows in set (0.00 sec)
```

Q5. List the ssn and full name of employees who have no dependents.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT e.ssn, e.fname, e.lname
    -> FROM EMPLOYEE e
    -> LEFT JOIN DEPENDENT d ON e.ssn = d.emp_ssn
    -> WHERE d.emp_ssn IS NULL;
+-----+-------+--------+
| ssn | fname | lname  |
+-----+-------+--------+
| 102 | Ramen | Kalita |
| 103 | Priya | Baruah |
| 105 | Deepa | Bora   |
+-----+-------+--------+
3 rows in set (0.00 sec)
```

Q6. List the ssn and full name of employees who work more than 10 hours per week on at least one project.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT DISTINCT e.ssn, e.fname, e.lname
    -> FROM EMPLOYEE e
    -> JOIN WORKS_ON w ON e.ssn = w.emp_ssn
    -> WHERE w.hours > 10;
+-----+--------+--------+
| ssn | fname  | lname  |
+-----+--------+--------+
| 100 | Arun   | Sarma  |
| 101 | Anjali | Das    |
| 102 | Ramen  | Kalita |
| 105 | Deepa  | Bora   |
+-----+--------+--------+
4 rows in set (0.00 sec)
```

Q7. List the ssn and full name of employees who work on at least one project that is NOT controlled by their own department.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT DISTINCT e.ssn, e.fname, e.lname
    -> FROM EMPLOYEE e
    -> JOIN WORKS_ON w ON e.ssn = w.emp_ssn
    -> JOIN PROJECT p ON w.proj_no = p.proj_no
    -> WHERE p.dept_no <> e.dept_no;
+-----+--------+-------+
| ssn | fname  | lname |
+-----+--------+-------+
| 101 | Anjali | Das   |
| 105 | Deepa  | Bora  |
+-----+--------+-------+
2 rows in set (0.00 sec)
```

Q8. Find the departments (dept_no, name) whose manager started before '2020-01-01'.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT dept_no, name
    -> FROM DEPARTMENT
    -> WHERE mgr_start_date < '2020-01-01';
+---------+----------------+
| dept_no | name           |
+---------+----------------+
|       1 | Research       |
|       2 | Administration |
|       3 | Headquarters   |
+---------+----------------+
3 rows in set (0.00 sec)
```

Q9. List projects (proj_no, name) that have the maximum total assigned hours across all employees (if multiple, list all).

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT w.proj_no, p.name
    -> FROM WORKS_ON w
    -> JOIN PROJECT p ON w.proj_no = p.proj_no
    -> GROUP BY w.proj_no, p.name
    -> HAVING SUM(w.hours) = (
    ->     SELECT MAX(total_hours)
    ->     FROM (
    ->         SELECT SUM(hours) AS total_hours
    ->         FROM WORKS_ON
    ->         GROUP BY proj_no
    ->     ) AS t
    -> );
+---------+----------+
| proj_no | name     |
+---------+----------+
|      10 | ProductX |
+---------+----------+
1 row in set (0.00 sec)
```

Q10. For each supervisor, list supervisor ssn, full name, and the number of direct reports they supervise.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT s.ssn AS supervisor_ssn, s.fname AS supervisor_fname, s.lname AS supervisor_lname,
    ->        COUNT(e.ssn) AS num_reports
    -> FROM EMPLOYEE s
    -> JOIN EMPLOYEE e ON e.super_ssn = s.ssn
    -> GROUP BY s.ssn, s.fname, s.lname;
+----------------+------------------+------------------+-------------+
| supervisor_ssn | supervisor_fname | supervisor_lname | num_reports |
+----------------+------------------+------------------+-------------+
|            100 | Arun             | Sarma            |           2 |
|            104 | Rajesh           | Gogoi            |           1 |
+----------------+------------------+------------------+-------------+
2 rows in set (0.00 sec)
```

Q11. For each project, display proj_no, project name, number of employees assigned, and total hours worked on that project.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT p.proj_no, p.name, COUNT(w.emp_ssn) AS num_employees,
    ->        COALESCE(SUM(w.hours), 0) AS total_hours
    -> FROM PROJECT p
    -> LEFT JOIN WORKS_ON w ON p.proj_no = w.proj_no
    -> GROUP BY p.proj_no, p.name;
+---------+-----------------+---------------+-------------+
| proj_no | name            | num_employees | total_hours |
+---------+-----------------+---------------+-------------+
|      10 | ProductX        |             3 |       47.00 |
|      20 | ProductY        |             1 |        5.00 |
|      30 | OfficeAutomation |             3 |       45.00 |
|      40 | HeadOfficeSystem |             2 |       15.00 |
+---------+-----------------+---------------+-------------+
4 rows in set (0.00 sec)
```

Q12. List the ssn and full name of employees who work on ALL projects controlled by their own department. (Hint: use NOT EXISTS with a nested NOT EXISTS.)

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT e.ssn, e.fname, e.lname
    -> FROM EMPLOYEE e
    -> WHERE NOT EXISTS (
    ->     SELECT *
    ->     FROM PROJECT p
    ->     WHERE p.dept_no = e.dept_no
    ->       AND NOT EXISTS (
    ->           SELECT *
    ->           FROM WORKS_ON w
    ->           WHERE w.emp_ssn = e.ssn
    ->             AND w.proj_no = p.proj_no
    ->       )
    -> );
+-----+--------+--------+
| ssn | fname  | lname  |
+-----+--------+--------+
| 100 | Arun   | Sarma  |
| 102 | Ramen  | Kalita |
| 103 | Priya  | Baruah |
| 104 | Rajesh | Gogoi  |
+-----+--------+--------+
4 rows in set (0.00 sec)
```

Q13. List employees (ssn, full name) whose salary is above the average salary of their department.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT e.ssn, e.fname, e.lname
    -> FROM EMPLOYEE e
    -> WHERE e.salary > (
    ->     SELECT AVG(e2.salary)
    ->     FROM EMPLOYEE e2
    ->     WHERE e2.dept_no = e.dept_no
    -> );
+-----+--------+-------+
| ssn | fname  | lname |
+-----+--------+-------+
| 100 | Arun   | Sarma |
| 104 | Rajesh | Gogoi |
+-----+--------+-------+
2 rows in set (0.00 sec)
```

Q14. List employees (ssn, full name) who have a dependent whose name matches the employee's first name (fname).

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT DISTINCT e.ssn, e.fname, e.lname
    -> FROM EMPLOYEE e
    -> JOIN DEPENDENT d ON e.ssn = d.emp_ssn
    -> WHERE d.name = e.fname;
+-----+--------+-------+
| ssn | fname  | lname |
+-----+--------+-------+
| 101 | Anjali | Das   |
+-----+--------+-------+
1 row in set (0.00 sec)
```

Q15. List the department(s) (dept_no, name) in which every employee earns strictly more than the average salary of all employees in the company.

Ans:

```
mysql> -- Student Name : Raktim Ranjan Handique
mysql> SELECT d.dept_no, d.name
    -> FROM DEPARTMENT d
    -> JOIN EMPLOYEE e ON d.dept_no = e.dept_no
    -> GROUP BY d.dept_no, d.name
    -> HAVING MIN(e.salary) > (
    ->     SELECT AVG(salary)
    ->     FROM EMPLOYEE
    -> );
+---------+--------------+
| dept_no | name         |
+---------+--------------+
|       3 | Headquarters |
+---------+--------------+
1 row in set (0.00 sec)
```